

Mechanics WS 4

- A particle of unit mass moves in a straight line against a resistance R given by $R = v(1 + v^2)$, where v is the velocity of the particle at a distance x metres from the origin. Prove that $x = \tan^{-1} v_0 - \tan^{-1} v$, where v_0 is the initial velocity at the origin. Use the formula $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left[\frac{A-B}{1+AB} \right]$ to show that: $v = \frac{v_0 - \tan x}{1 + v_0 \tan x}$
- A particle of unit mass travels in a straight line against the resisting force $f(v) = v(1 + v^2)$. Its initial velocity is c m/s at the origin. Show that the time t , when velocity is v , is given by $t = \frac{1}{2} \log \left[\frac{1+v^2}{1+c^2} \right]$. Find v^2 as a function of t and hence the limiting value of v at $t \rightarrow \infty$.
- A particle of mass m moves along a straight line under the action of a constant (propelling) force P , and a resistive force mkv , where k is a constant, v is the speed at any time t . Show that if the speed increases from 2 m/s to 4 m/s over a time interval of 5 seconds,
 - $P = 2km \left[\frac{2e^{5k} - 1}{e^{5k} - 1} \right]$.
 - Find the corresponding distance moved.
 - Find the propelling force P for $k = 0.5$
- The acceleration due to gravity at a distance r from the earth's centre is directed towards the centre and equal to $\frac{k}{r^2}$ when $r > R$, equal to kr when $r < R$ and equal to g when $r = R$. Imagine a narrow tunnel along a diameter XY of the earth and the particle is projected from X with initial velocity U towards Y . ($R = \text{radius of earth}$). If $U^2 < 2gR$, prove that the motion is oscillatory and the amplitude is given by $\frac{R}{1 - \frac{U^2}{2gR}}$.
- A body of mass m is released from a height h above the ground and it experiences a resistive force R given by $R = 0.1mv^2$ where v is the velocity achieved by the body at time t . If the object falls from rest under gravity ($g = 10 \text{ m.s}^{-2}$). Find:
 - the terminal velocity U (i.e. velocity as $t \rightarrow \infty$)
 - the height h if the velocity is $0.5U$ just before the body strikes the ground.
- A particle P of unit mass is projected from a point O with velocity U at an angle θ to the horizontal. The particle moves under gravity and each component of its velocity experiences a resistance k times the magnitude of the component. By considering the rectangular axes Ox and Oy and the particle has components v_x and v_y of the velocity v at time t :
 - Draw a diagram of the motion of P ,
 - Write down the equations of motion in the forms:

$$\frac{d}{dt}(v_x) = \dots \quad \text{and} \quad \frac{d}{dt}(v_y) = \dots$$
 - Prove that $v_x = U \cos \theta e^{-kt}$ and $v_y = \frac{1}{k}(g + kU \sin \theta)e^{-kt} - \frac{g}{k}$
 - Prove: $x = \frac{U}{k} \cos \theta (1 - e^{-kt})$, $y = \frac{(kU \sin \theta + g)}{k^2} (1 - e^{-kt}) - \frac{gt}{k}$.
- A particle is projected vertically upwards from the surface of the earth with initial velocity U . The acceleration due to gravity at a distance x from the centre of the earth is given by $\frac{k}{x^2}$, directed towards the centre. Prove that the rocket will escape from the earth provided $U^2 \geq 2gR$, where g is the acceleration due to gravity at earth's surface and R is the earth's radius. Further, if $U^2 = 2gR$, show that the time to achieve the height R above the earth's surface is approximately equal to $0.273\sqrt{R}$, given $g = 10 \text{ m.s}^{-2}$.
- A projectile is fired vertically upwards from the earth's surface with velocity U m/s. The retardation due to gravity is given by the law $\frac{k}{x^2}$ where x is the distance of the projectile from the centre of the earth, and k is a constant. The acceleration due to gravity on the earth's surface is g . The earth's radius

is R . Neglecting the air resistance, show that if $U^2 = gR$, then the projectile reaches the height R above the earth's surface. Show further that the time for this journey

$$\text{is } \left[\frac{\pi}{2} + 1\right] \sqrt{\frac{R}{g}}.$$

9. The rise and fall of the tide at the mouth of the Ord River in Western Australia is assumed to be Simple Harmonic Motion. The depth of water at low tide on a particular day was 0.7 metres and the depth of water at high tide was 3.7 metres. If low tide occurred at 8:55am and high tide at 3:05pm, find the earliest possible time after low tide at which a fishing boat could enter the mouth of the river if it required the water to be at least 2 metres deep.
10. In an experiment, water in a tank rises and falls with simple harmonic motion. The greatest depth of water is 9 metres and the least depth is 1 metre. At 7am the depth of water was 5 metres and increasing. Three hours later the depth has reached 9 metres for the first time. Given that the depth, x metres, of water time t hours after 7am can be represented by the formula $x = b + a \sin nt$, find:
 - a. The values of b , a and n
 - b. The time after 7am when the depth of water first reaches 3 metres.
11. A particle moving in simple harmonic motion, has displacement, y metres from the origin

such that $y = \sin(2t - \frac{\pi}{4}) + \sin(2t)$. Find the amplitude of this motion.

12. The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = 2x - 4$ where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is at rest at $x = 4$.
 - a. If the velocity of the particle is v m/s, show that $v^2 = 2(x^2 - 4x)$
 - b. Show that the particle does not pass through the origin
 - c. Find the position of the particle when $v = \sqrt{90}$. Justify your answer
13. Find the time taken for the particle to slow down to a speed of 1% of its initial speed. The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = (4x - 4)$, where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is 6 metres to the right of O and its velocity (v m/s) is -8 m/s.
 - a. Show that $v^2 = 4x^2 - 8x - 32$.
 - b. Find the set of possible values of x where motion can exist and describe the motion of the particle.
14. The velocity v of a body is given by $v = x \tan^2 x$, where x is the displacement. Find in simplest terms the acceleration of the body in terms of x .

Answer:

- 1.
2. $v^2 = \frac{1}{(1+c^{-2})e^{2t}-1}, 0$
3. (b) $\frac{1}{K}[-2 + \frac{p}{mk} \log(\frac{p-2mk}{p-4mk})]$ (c) 2.1 m N
- 4.
5. (a) 10 m/s (b) 1.44 m
6. (b) $-kv_x$ and $-g - kv_y$
- 7.
- 8.
9. 11:44am

$$10. \text{ a. } b = 5, a = 4, n = \frac{\pi}{6}; \text{ b. } 2\pi\text{m}$$

$$11. A = \sqrt{2 + \sqrt{2}}$$

$$12. \text{ c. } x = 9$$

$$13. \text{ b. } x \geq 4 \text{ or } x \leq -2$$

$$14. x \tan^3 x (\tan x + 2x \tan^2 x)$$